

Sukirat Kaur

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

La Jolla, CA
Expected 2029

EXPERIENCE

Tech Director

DriveSmartVE

Sep. 2024 – June 2025

Virtual Enterprise Program

- Led the design and development of an e-commerce website for a startup simulation using Wix.
- Directed technical decisions and coordinated development efforts across the team.
- Project earned 1st Place at the San Diego Virtual Enterprise Conference and placed in the Top 10% nationally for design and functionality.

PROJECTS

Arduino Circadium Rhythm House | *Arduino, Embedded Systems*

HardHacks UCSD 2026

- Built a rhythm-based house project using Arduino that matched interior lighting to the lighting patterns outside the house.
- Integrated hardware and software components to synchronize lighting behavior in real time.
- Collaborated in a hackathon setting to quickly prototype and present a working embedded systems project.

Fruit Face Match | *Python, Raspberry Pi, Machine Learning*

Aug. 2025

- Developed a machine learning model using Raspberry Pi that detects human faces and audibly announces the fruit they most resemble.
- Combined computer vision and audio output to create an interactive real-time system.

Entropy in Machine Learning | *Machine Learning, Physics*

2024 – 2025

- Conducted independent research exploring the relationship between entropy and machine learning efficiency.
- Developed and tested models analyzing how disorder and information theory influence algorithmic performance.
- Earned recognition at the Kern County Science Fair for applying physics concepts to computer science.

Healthy Recipe Recommender | *Python*

May 2025

- Designed and implemented a Python-based recipe recommender that suggests nutritious meals tailored to users' health conditions.
- Built logic for personalized recommendations based on user needs.

TECHNICAL SKILLS

Languages: Java, Python, C

Tools: Raspberry Pi, Arduino

Relevant Coursework: Accelerated Introduction to Programming, Basic Data Structures & Object-Oriented Design, Discrete Mathematics, Systems Programming and Software Tools

AWARDS

Top 10% National — E-Commerce Website

Kern County Science Fair 3rd Place — Entropy in Machine Learning

SPIS Certificate of Completion